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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCHSC202 –

FOOD SCIENCE AND FOOD SAFETY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Name the building blocks of proteins. [K] / [CO1]
   - Amino Acids (2marks)
2. How does low temperature help in preserving food? [K] / [CO4]
   - Slow down the growth of microorganisms (1mark)
   - Reduce enzyme and chemical activities (1mark)
3. Name two personal hygiene practices important for food handlers. [K] / [CO5]
   - Maintain proper hand hygiene and wear clean protective clothing. (1mark)
   - Avoid handling food when sick and keep nails, cuts, and body clean(1mark)
4. Give two examples of protective foods in My Healthy Plate. [K] / [CO1]
   - Fruits / Vegetables (2marks)
5. What is pickling in food preservation? [K] / [CO4]

- Pickling is a method of food preservation in which food is preserved using salt, vinegar, or oil, which creates an acidic environment that prevents the growth of microorganisms. (2marks)

6. Name one organization responsible for food quality control in India. [K] / [CO5]
   - **FSSAI** - Food Safety and Standards Authority of India.

7. How does food provide comfort and satisfaction? [U] / [CO1]
   - By fulfilling hunger (1mark)
   - By giving pleasure through taste, aroma, and appearance, which also has a positive effect on emotions (1mark)

8. How does preservation help in reducing food wastage? [K] / [CO4]
   - By preventing spoilage and extending the shelf life of food (2marks)

9. Why is pH an important chemical parameter in food quality control? [K] / [CO5]
   - Because it affects microbial growth and food safety (1mark)
   - Influences taste, texture, and shelf life of food. (1mark)

10. Give two examples of pulses included in the food group system. [K] / [CO1]
    - Lentils / Chickpeas / Green gram /Black gram [1mark for each (any two options)]

11. Give two examples of body-building foods. [K] / [CO1]
    - Milk /Eggs / Fish /Pulses /Meat [1mark for each (any two options)]

12. How does high temperature help in preserving food? [K] / [CO4]
    - By destroying microorganisms (1mark)
    - By inactivating enzymes (1mark)

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

13.Describe the role of hygiene and cleanliness in food preparation, cooking, and serving to ensure safe food. [U] / [CO5]

(1 mark each)

- Prevents contamination
- Ensures safe cooking
- Maintains food quality
- Prevents foodborne diseases

14.Classify vegetables with examples. [K] / [CO2]

(1 mark each)
- Root vegetables – Carrot, Beetroot
- Stem vegetables – Potato, Ginger
- Leafy vegetables – Spinach, Cabbage
- Flower vegetables – Cauliflower, Broccoli
- Fruit vegetables – Tomato, Brinjal

15.Given a scenario of storing leftover cooked rice, recommend a method to preserve it safely and justify your choice. [A]/ [CO4]

- Store the leftover cooked rice by refrigeration (below 4°C). (2marks)
- **Justification**: Low temperature slows the growth of microorganisms (especially harmful bacteria) and prevents spoilage, making the rice safe for later consumption. (2marks)

16.Describe the nutritive value of pulses. [K] / [CO2]
- **Protein** (20–25 g): Excellent source of plant protein, but deficient in methionine. (1mark)
- **Fat** (1–2 g): Low in fat; exceptions like Soybean and Groundnut contain higher fat. (½ mark)
- **Carbohydrate**s (55–60 g): Main component, mostly starch with some fibre. (1mark)
- **Fibre** (1–5 g): Good source of dietary fibre, especially when whole. (½ mark)
- **B Vitamins**: Provide thiamine, riboflavin, niacin, and folate. (½ mark)
- Rich in **iron** and **Calcium** (½ mark)

17.Explain the nutritive composition of nuts. [K] / [CO3]

- **Energy** (550–700 kcal): Highly energy-dense. (½ mark)
- **Protein (**15–25 g): Good plant protein source. (1 mark)
- **Fa**t (45–70 g): Rich in healthy fats; higher saturated fat in Macadamia. (½ mark)
- **Carbohydrates** (15–30 g) & **Fibre** (8–12 g): Moderate carbs and good fibre. (1 mark)
- Rich in **Vitamin E** (e.g., Almonds), **B-complex vitamins** (½ mark)
- Contains minerals like **magnesium, iron, zinc**; high selenium in Brazil nuts (½ mark)

18.Discuss the importance of temperature control in food safety for perishable foods. [U] / [CO5]

- Temperature control is important in food safety for perishable foods because it prevents

the rapid growth of microorganisms that cause spoilage and foodborne illnesses. (2marks)

- Keeping foods at safe temperatures (refrigeration or proper cooking) also slows enzyme activity and maintains quality, thereby extending shelf life and ensuring safety. (2marks)

19.Discuss the major classes of nutrients and state their primary functions. [U] / [CO1]

- **Carbohydrates:** Main source of energy for the body. (1 mark)
- **Proteins:** Body building; growth and repair of tissues. (1 mark)
- **Fats:** Concentrated source of energy and help in absorption of fat-soluble vitamins. (1 mark)
- **Vitamins:** Regulate body processes and support immunity. (½ mark)
- **Minerals:** Help in body structure (bones, teeth) and regulate functions. (½ mark)

[4x5 = 20]

Part C

Essay Questions
Answer any **1** questions
Each question carries **10** marks

20.Compare preservation by osmosis with high or low temperature, highlighting advantages, limitations, and suitable foods for each method. [A]/ [CO4]

(2 marks each for each aspects)

Aspect	Osmosis (salting/sugaring)	High Temperature	Low Temperature
Definition	Preservation by using salt or sugar to draw out water from food and microbes	Preservation by applying heat to destroy microorganisms and enzymes	Preservation by using low temperatures to slow microbial and enzyme activity
Principle	Removes water from food and microbes by osmotic pressure	Destroys microorganisms and inactivates enzymes	Slows microbial growth and enzyme activity
Advantages	Simple, inexpensive, improves taste, no refrigeration needed	Ensures safety, increases shelf life	Preserves freshness and nutrients
Limitations	Alters taste/texture; not suitable for all foods	Loss of some nutrients; may affect flavour	Needs electricity; limited shelf life in refrigeration
Suitable Foods	Pickles, jams, jellies, salted fish	Milk, canned foods, cooked foods	Fruits, vegetables, meat, leftovers

21.Classify milk and milk products. Explain their nutritive composition and importance in growth and development. [K] / [CO3]

Classification of Milk and Milk Products (4 marks)

A. Based on Fat Content / Standardization

- Whole milk: ~3.5–6% fat
- Toned milk: ~3% fat
- Double-toned milk: ~1.5% fat
- Skimmed milk: <0.5% fat
- Full-cream milk***:*** ~6% fat

B. Based on Processing

- Pasteurized milk: Heated to destroy pathogens
- Sterilized milk: Longer shelf life, slight nutrient loss
- UHT milk: Long shelf life, minimal nutrient loss
- Homogenized milk: Uniform texture, no cream separation
- Reconstituted milk: Made from milk powder
- Concentrated products: Condensed milk, evaporated milk

C. Fermented Milk Products

- Curd, yoghurt, buttermilk, kefir

Nutritive Composition of Milk (per 100 ml) (4 marks)

- Energy: 60–70 kcal
- Protein: ~3.2 g (high biological value)
- Fat: 3.5–4 g
- Carbohydrates: ~4.8 g (lactose)
- Minerals: Calcium (120 mg), phosphorus, potassium
- Vitamins: A, D, B2, B12
- Water: 85–88%

Importance in Growth and Development (2 marks)

- Promotes bone and teeth formation (rich in calcium & phosphorus)
- Supports growth and tissue repair (high-quality protein)
- Important for children, adolescents, and pregnant women
- Helps in overall body development and maintenance

[10x1 = 10]